

Unit-3

PERIODIC PROPERTIES

NOTE: 1-2 block 3-12 block 13-18 block f-block
s-block d-block p-block (inner transition element)

GENERAL ELECTRONIC CONFIGURATION →

Group 1 → (Alkali metals) → ns^1 } s block

Group 2 → (Alkaline earth metal) → ns^2 }

For p-Block ELEMENT → $ns^2 np^{1-6}$

For e.g. → Group 13 → $ns^2 np^1$ Group 16 → $ns^2 np^4$

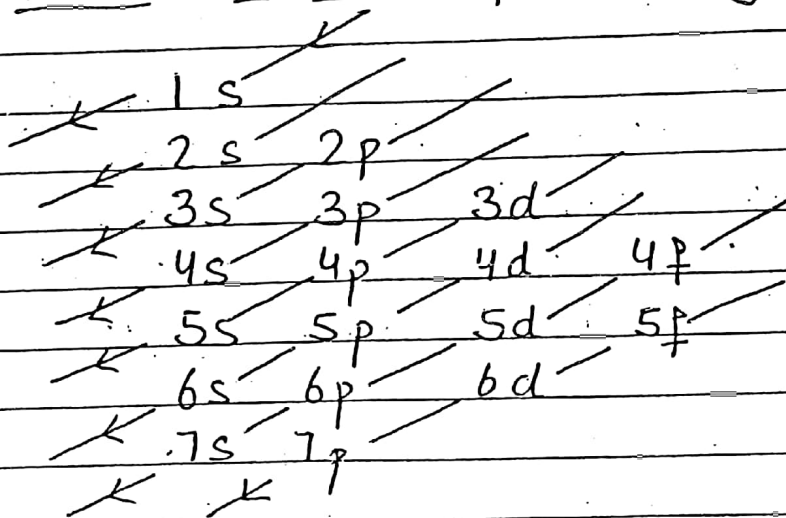
Group 14 → $ns^2 np^2$ Group 17 → $ns^2 np^5$

Group 15 → $ns^2 np^3$ Group 18 → $ns^2 np^6$

FOR d-BLOCK ELEMENT → $(n-1)d^{1-10} ns^{0-2}$

FOR f-BLOCK ELEMENT → $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$

to ~~write~~ → orbitals in order of increasing order energy →



1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f
6p 7s 5f 6d 7p

s p d f

(1) Angular Momentum 0 1 2 3

for 1s → $n=1$, $l=0$, $n+l=1+0=1$

for 2s → $n=2$, $l=0$, $n+l=2+0=2$

For 4s $\rightarrow n=4, l=0$ for 3d $\rightarrow n=3, l=2$
 $n+l = 4+0 = 4$ $n+l = 3+2 = 5$

NOTE $\rightarrow$ lower $n+l$ value is filled first that's why we filled 4s first then we filled 3d.

For 3d $\rightarrow n=3, l=2$ for 4p $\rightarrow n=4, l=1$
 $n+l = 3+2 = 5$ $n+l = 4+1 = 5$

NOTE $\rightarrow$ If value of $n+l$ in both cases is same then we see value of n . lower n value is filled first that's why we filled 3d first then we filled 4p.

ELECTRONIC CONFIGURATION $\rightarrow$ Ar $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6$

K $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

Sc $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$

Ti $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$

V $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$

✓ Cr $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

($\because$ half filled orbitals gains stability)

✓ Cu $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

($\because$ fully filled orbitals gains stability)

In case of Ions $\rightarrow$ Cu⁺ $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{10}$

($\because$ Total effective nuclear charge of ~~4s~~ ^{3d} is higher, So we remove $\bar{e}$ from 4s)

Cu²⁺ $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$

Fe $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

Fe³⁺ $\rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

EFFECTIVE NUCLEAR CHARGES $\rightarrow$

$\rightarrow$ Effective Nuclear charge is the reduce nuclear charge felt by an $\bar{e}$ due to screening effect.

→ Screening effect is the repulsive effect b/w the e^- present in inner shell or the same shell

$$Z_{\text{eff}} = Z - \sigma$$

Whereas

$Z_{\text{eff}} \rightarrow$ effective Nuclear charge

$Z =$ original Nuclear charge

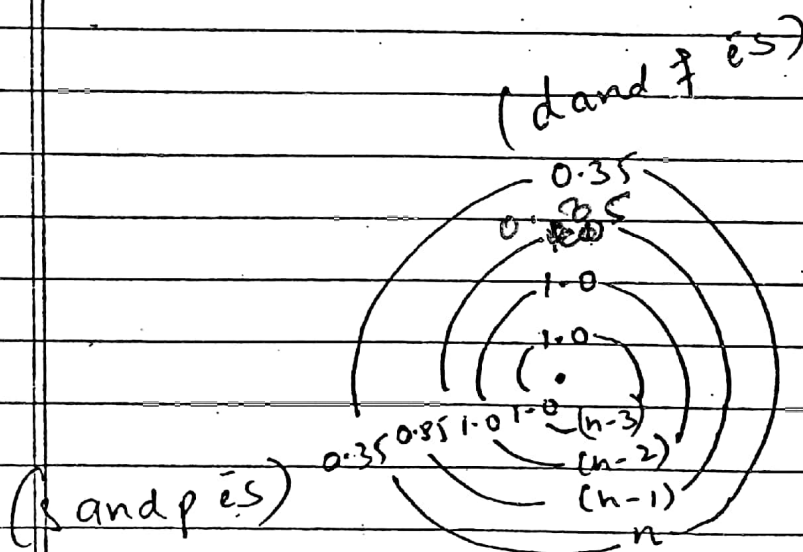
$\sigma =$ Screening Constant

SLATER'S RULE (To CALCULATE σ) $\rightarrow$

Steps of Slater's Rule $\rightarrow$

Step 1. $\rightarrow$ Arrange the orbitals in group
(1s) (2s 2p) (3s 3p) (3d) (4s 4p) (4d) (4f) (5s 5p)

Step 2. $\rightarrow$ Contribution per e^- in an orbital



examples $\rightarrow$

(i) σ for 2p e^- in N (n-1) n
 $N \rightarrow (1s)^2 (2s)^2 (2p^3) = (1s)^2 (2s^2 2p^3)$

$$Z_{\text{eff}} = 7 - \sigma = 7 - 3.1 = 3.9$$

$$\sigma = 4 \times 0.35 + 0.85 \times 2 = 3.1$$

(ii) σ for $2s$ $\bar{e}$ in N

Ans

$$\sigma = 1 \times 0.35 + 2 \times 0.85 = 2.05$$

$$Z_{eff} = Z - \sigma$$

$$Z_{eff} = 7 - 2.05 = 5.0$$

(iii)
Ans

σ for $3s$ $\bar{e}$ of Na $(n-2)$ $(n-1)$ n
As Na = 11 $\rightarrow 1s^2 (2s^2 2p^6) (3s^1)$

$$Z_{eff} = 11 - \sigma$$

$$\sigma = 0 \times 0.35 + 8 \times 0.85 + 2 \times \overset{1.0}{0.35} = 8.8$$

$$Z_{eff} = 11 - 8.8 = 2.2$$

(iv)
Ans

σ for $4s$ $\bar{e}$ of Cr
 $Cr = \underbrace{(1s^2)}_{(n-3)} \underbrace{(2s^2 2p^6)}_{(n-2)} \underbrace{(3s^2 3p^6)}_{(n-1)} \underbrace{(3d^5)}_{(n-1)} \underbrace{(4s^1)}_n$

$$Z_{eff} = 24 - \sigma$$

$$\sigma = 0 \times 0.35 + 0.85 \times 13 + 8 \times 1.0 + 2 \times 1.0 = 21.05$$

$$Z_{eff} = 24 - 21.05 = 3$$

(v)
Ans

σ for $3d$ $\bar{e}$ of Cr
 $Cr = \underbrace{(1s^2)}_{(n-2)} \underbrace{(2s^2 2p^6)}_{(n-1)} \underbrace{(3s^2 3p^6)}_{(n)} \underbrace{(3d^5)}_{(n)} \underbrace{(4s^1)}_{(X) \text{ (not contribut)}}$

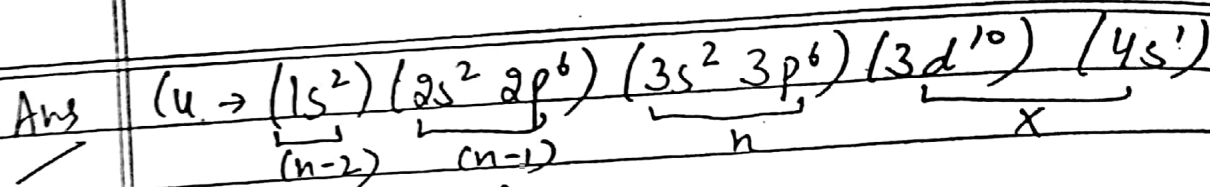
$$Z_{eff} = 24 - \sigma$$

$$\sigma = 12 \times 0.35 + 8 \times 1.0 + 2 \times 1.0 = 14.2$$

$$Z_{eff} = 24 - 14.2 = 9.8$$

(vi)

σ for $3p$ $\bar{e}$ of Cr



$$Z_{eff} = 29 - \sigma$$

$$\sigma = 7 \times 0.35 + 0.85 \times 8 + 2 \times 1.0 = 11.25$$

$$Z_{eff} = 29 - 11.25 = 17.75$$

NOTE $\rightarrow$ If you are ask to find σ for $1s \bar{e}$, the contribution per $\bar{e}$ is 0.30

orbital	Contribution per $\bar{e}$			
	n	(n-1)	(n-2)	(n-3)
s and p	0.35	0.85	1.0	1.0
d and f	0.35	1.0	1.0	1.0

Penetration of orbitals $\rightarrow$

Penetration $\rightarrow$ It is the tendency of $\bar{e}$ to go towards the nucleus.

Penetration of orbitals $\rightarrow$ It is the tendency of cloud to get attracted towards the nucleus.

$\rightarrow$ The Penetration depends upon the no. of elec and shape of electron. (due to uniform shape)

For same shell penetration is $s > p > d > f$

For different shell penetration is

$1s > 2s > 2p > 3s > 3p > 4s > 3d > 5s > 6p$

ATOMIC RADIUS AND IONIC RADIUS $\rightarrow$

ATOMIC RADIUS $\rightarrow$ Atomic Radius is defined as the distance from

centre of nuclei to the outermost shell where the e^- are present.

→ Atomic Radii is determined by X-ray / spectral studies. 

VARIATION OF ATOMIC RADII IN PERIODIC TABLE →

(a) In Periods → Decreases from left to right

As we move from left to right in periodic table, No. of shell remain constant (n) but no. of e^- increases. As Nuclear charge (Z) increase due to which force of attraction of nucleus for e^- s increases. As force of attraction inc., Atomic radius (r) decreases.

→ No. of shell (n) remains constant → No. of e^- s increases → Nuclear charge (Z) increases → Force of attraction of nucleus for e^- s increases (F.A.) → Atomic radius (r) decreases.

(b) In Group (from top to bottom) →

effect (i) no. of shell increases → force of attraction dec. → Atomic radii increases.

Also, effect (ii) no. of electron inc. → Nuclear charge increases → radius dec.

(i) effect > 2nd effect

But the effect of inc. of no. of shell is more than the effect of inc. in nuclear charge. So, [REDACTED]

Atomic radius from top to bottom

IONIC RADII → It is the radius of ion in the ionic crystal is known as ionic radii.

- Ions is formed due to loss or gain of electron. If e^- s are lost then cation is formed. If e^- s are gained then anion is formed.
- Ionic radii is also determined by X-ray / Spectra studies.

SIZE OF CATION → size of cation is smaller than an atom.

This is because the in case of cation, no. of e^- dec but nuclear charge remain constant. If nuclear charge remain constant then effective of nuclear charge per e^- inc. acc. to formula ($Z_{eff} = Z - \sigma$). If effective nuclear charge per e^- inc., then, force of attraction increase, due to which size of cation dec.

→ $e^- \downarrow \rightarrow Z \text{ constant} \rightarrow Z_{eff} \text{ per } e^- \uparrow \rightarrow F.A \uparrow \rightarrow r \downarrow$

SIZE OF ANION → size of Anion is larger than an atom.

This is because in case of anion, no. of e^- s inc. but nuclear charge remain constant. If nuclear charge remain constant then effective nuclear charge per e^- dec. acc. to formula ($Z_{eff} = Z - \sigma$). If effective nuclear charge per e^- dec, then force of attraction also dec. due to which size of anion inc.

→ $r_s \uparrow \rightarrow Z \text{ constant} \rightarrow Z_{eff} \downarrow \rightarrow F.A \downarrow \rightarrow r \uparrow$

VARIATION OF IONS IN PERIODIC TABLE

(a) In periods $\rightarrow$ Size of cation and anion dec. from left to right.
 For e.g $\rightarrow Li^+ > Be^+ > B^+ > C^+ > N^+ > O^+$

(b) In Group $\rightarrow$ Size of cation and anion $\uparrow$ from top to bottom.
 For e.g $\rightarrow Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

Q $\rightarrow C^{4-}, N^{3-}, O^{2-}, F^-$ in which more is greater size?
 $\rightarrow \searrow$

Ans $\rightarrow C^{4-} > N^{3-} > O^{2-} > F^-$

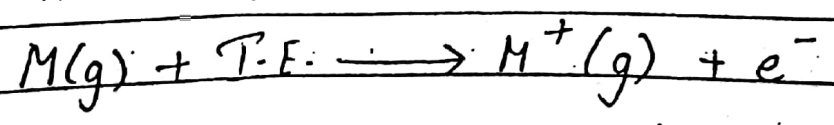
This is isoelectronic species

	C^{4-}	N^{3-}	O^{2-}	F^-
e^-	10	10	10	10
Z	6	7	8	9

As Z_{eff} is more for F^- So, size is less for F^- .

e^- all same $\leftarrow e^- \rightarrow Z \uparrow \rightarrow Z_{eff} \uparrow \rightarrow F.A \uparrow \rightarrow r \downarrow$
 # IONIZATION ENERGY $\rightarrow$ (I.E.) $\rightarrow$

It is the amount of energy require to remove an electron from outermost shell of 1 mole of isolated gaseous atom.



$\rightarrow$ If I.E. is the smaller, then it is easier to

remove the $\bar{e}$ s. But if I.E. is larger, then difficult to remove the $\bar{e}$ s.

FACTOR AFFECTING I.E. $\rightarrow$

1) Size of an atom (r) $\rightarrow$

If Size of atom (r) $\uparrow \rightarrow$ Force of attraction $\downarrow$
I.E. $\downarrow$

2) Nuclear charge (Z) $\rightarrow$

If Nuclear charge (Z) $\uparrow \rightarrow$ Force of attraction $\uparrow$
I.E. $\uparrow$

3) Screening Effect (σ) $\rightarrow$

If Screening effect (σ) $\uparrow \rightarrow Z_{eff} \downarrow$ (acc. to formula $Z_{eff} = Z - \sigma$) $\rightarrow$ Force of attraction $\downarrow \rightarrow$ I.E. $\downarrow$

4) Penetration of $e^- \rightarrow$

If Penetration $\uparrow \rightarrow \bar{e}$ s are closer to nucleus $\rightarrow$ Force of attraction $\uparrow \rightarrow$ I.E. $\uparrow$

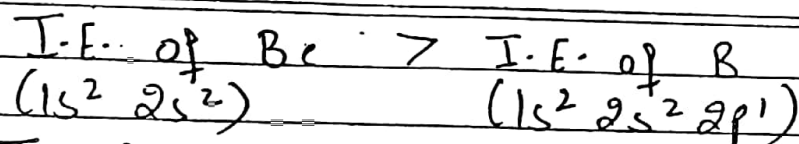
5) Electronic configuration $\rightarrow$

For half filled or fully filled orbitals $\rightarrow$ Stable $\rightarrow$ difficult to remove $\bar{e} \rightarrow$ I.E. $\uparrow$

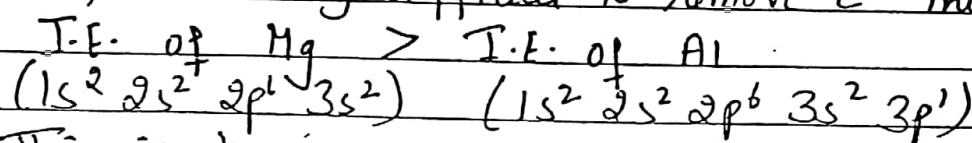
Q $\rightarrow$ Why I.E. of N is more than O?

Ans $\rightarrow$ I.E. of N $>$ I.E. of O
($1s^2 2s^2 2p^3$) ($1s^2 2s^2 2p^4$)

This is bcz, The electronic configuration of N is half filled. So, it is difficult to remove $\bar{e}$ from N. Hence, I.E. of N is $>$ than O.



This is bcz, In Be, s orbital is fully filled. So, it is very difficult to remove e^- . Thus, I.E. $\uparrow$



This is bcz, In Mg, s orbital is fully filled and closer to Nucleus. So, difficult to remove e^- . Thus, I.E. $\uparrow$.

Variation of I.E. In Periodic Table $\rightarrow$

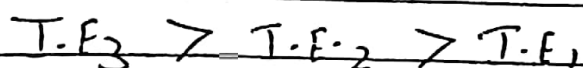
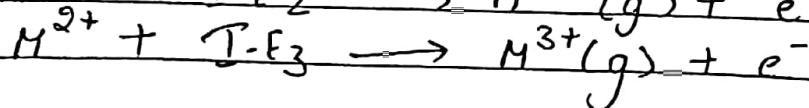
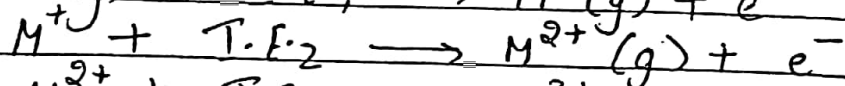
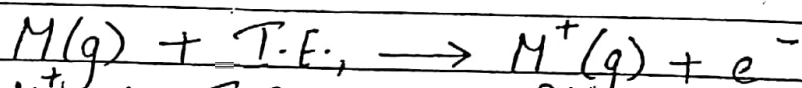
(a) In periods (from left to Right) $\rightarrow$

no. of shell remain constant $\rightarrow$ no. of e^- $\uparrow$ $\rightarrow$
 Nuclear charge (Z) $\uparrow$ $\rightarrow$ Force of attraction $\uparrow$ $\rightarrow$
 I.E. $\uparrow$

(b) In group (from Top to bottom) $\rightarrow$

Overall n $\uparrow$ $\rightarrow$ Force of attraction $\downarrow$ $\rightarrow$ I.E. $\downarrow$

SUCCESSIVE IONIZATION ENERGY $\rightarrow$



is because

As, charge of proton inc. i.e. e^- remove inc, then size decreases due to which force of

attraction of nucleus for e^- inc. and Ionization energy inc.

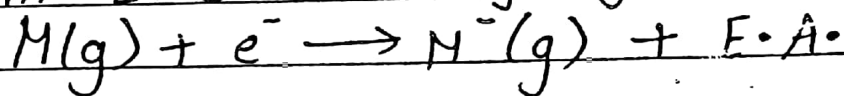
→ Charge $\uparrow \rightarrow r \downarrow \rightarrow F \cdot A \uparrow \rightarrow I.E. \uparrow$

Q → Which has Li Be B C N O F Ne has greater Ionization energy?

Ans → $Li < B < Be < C < O < N < F < Ne$

ELECTRON AFFINITY → (E.A) →

It is the amount of energy released when an e^- is added to the outermost shell of 1 mole of isolated atom. It is the tendency to gain the e^- .



Factors Affecting E.A →

(i) Size of atom (r) →

If size of atom (r) $\uparrow \rightarrow$ force of attraction $E.A \downarrow$

(ii) charge on Nucleus (Z) →

If $Z \uparrow \rightarrow F.A. \uparrow \rightarrow E.A. \uparrow$

(iii) Screening effect (σ) →

If $\sigma \uparrow \rightarrow Z_{eff} \downarrow \rightarrow F.A \downarrow \rightarrow E.A \downarrow$

(iv) Penetration of e^- →

If Penetration of $e^- \uparrow \rightarrow F.A. \uparrow \rightarrow E.A.$

(v) Electronic configuration →

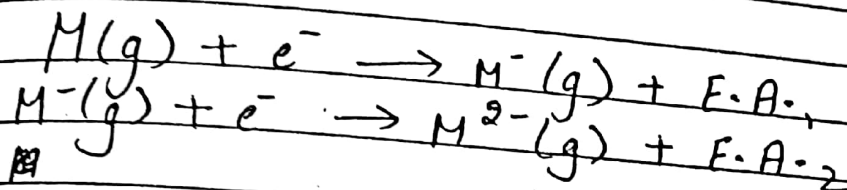
If F.C. is stable → Nearly zero E.A.

(half filled or fully filled orbital)

Variation of E.A. in Periodic Table →

- (a) In Periods (from left to right) →
If $\bar{e} \uparrow \rightarrow Z \uparrow \rightarrow F.A \uparrow \rightarrow E.A \uparrow$
- (b) In Group (from Top to bottom) →
Overall $r \uparrow \rightarrow F.A \downarrow \rightarrow E.A \downarrow$

SUCCESSIVE E.A. →



$$E.A._2 > E.A._1$$

It is difficult to add an $\bar{e}$ to an anion due to its large size and coulombic repulsion b/w negatively charged anion and the $\bar{e}$ to be added.

→ Halogens are ^{very} higher electron affinity bec to complete its octet it requires only the ~~from~~ one $\bar{e}$.

* MOLECULAR GEOMETRIES →

→ VSEPR Theory (Valence shell electron pair repulsion Theory)

Repulsion Order →

lone pair - lone pair repulsion > lone pair - bond pair repulsion > bond pair - bond pair repulsion

→ Repulsion inc.

← Angle dec.

→ When repulsion inc. then angle dec.

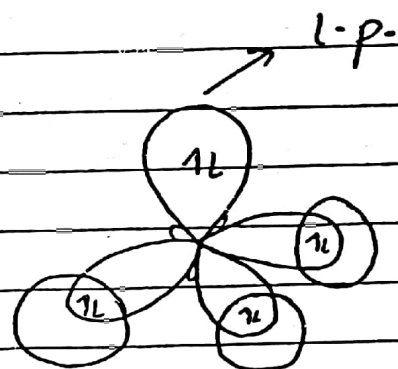
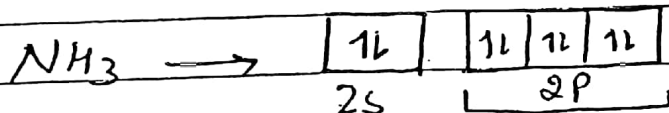
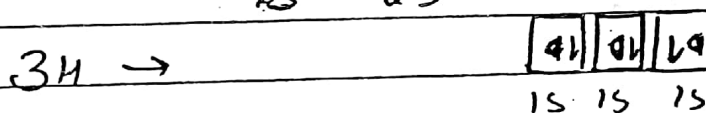
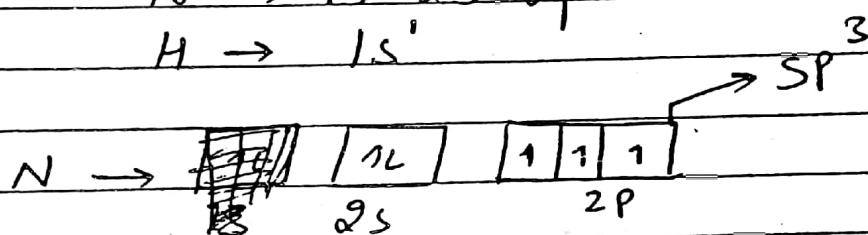
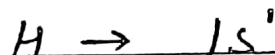
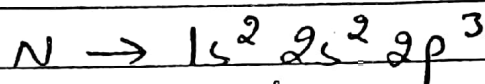
~~In presence of lone pair, the geometry change~~
~~For e.g. → if shape~~

→ Valence shell e take part in bond formation.

→ If atom is surrounded by bond pair in regular way, then geometry is regular.

→ If atom is surrounded by lone pair and bond pair, then geometry is irregular.

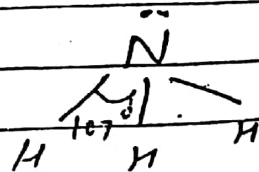
(1) NH_3



$\therefore 4 \text{ sp}^3$ hybridisation
l.p. b.p. Tetrahedral geometry

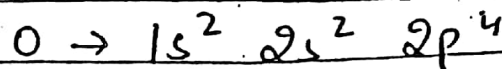
Shape $\rightarrow$ Trigonal pyramidal

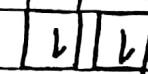
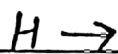
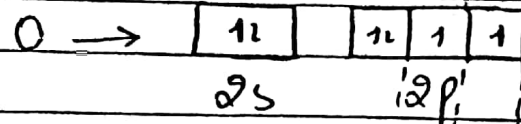
Due to repulsion due to presence of lone pair, angle dec. from 109° to 107°



2)

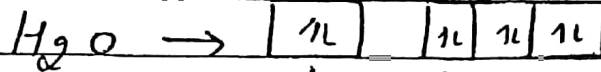
H_2O





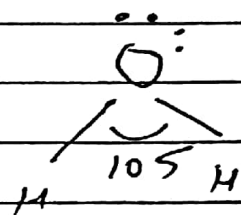
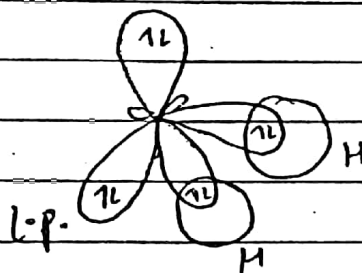
Bond pair

sp^3 hybridisation



lone pair

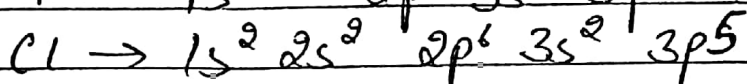
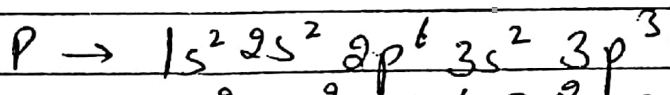
l.p.



Geometry Tetrahedral

Due to presence of Two lone pair, the Shape is bent and angle dec.

3) PCl₅

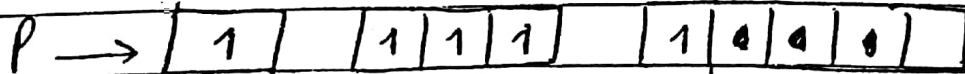


3s

3p

(Ground state)

In presence of Cl, one e from 3s excite to 3d,

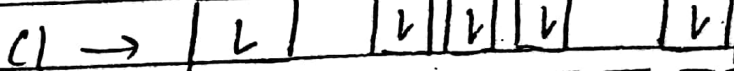


3s

3p

3d

sp^3d hybridisation



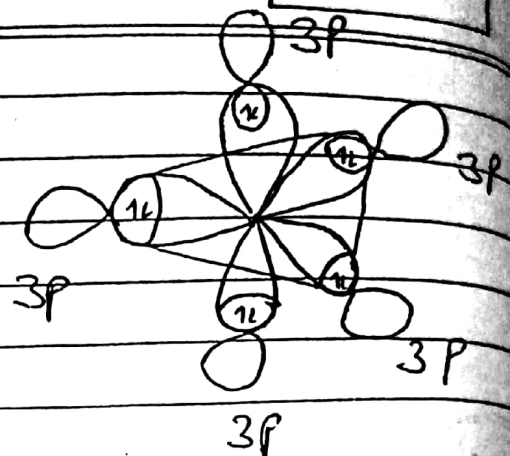
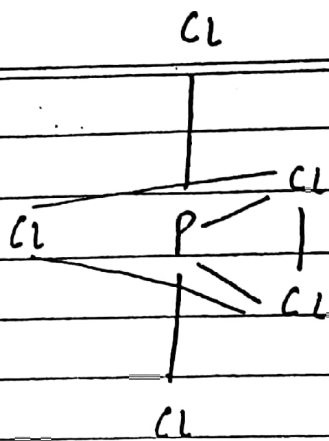
3p

3p

3p

3p

3p

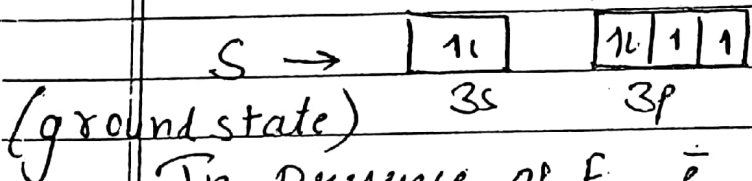
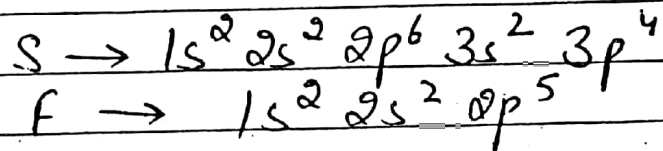


Shape $\rightarrow$ Regular Trigonal bipyramidal

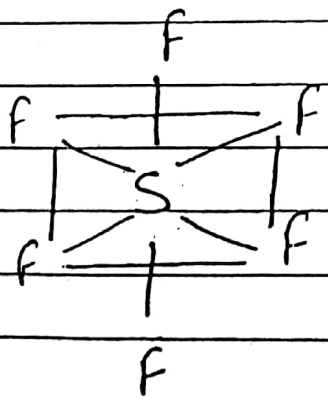
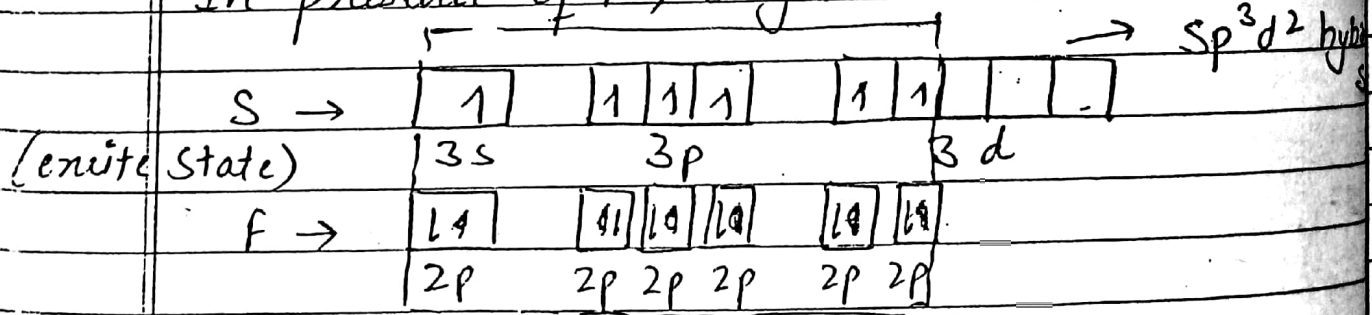
Geometry $\rightarrow$ Trigonal bipyram

$\rightarrow$ Due to absence of lone pair, the shape and geometry is same.

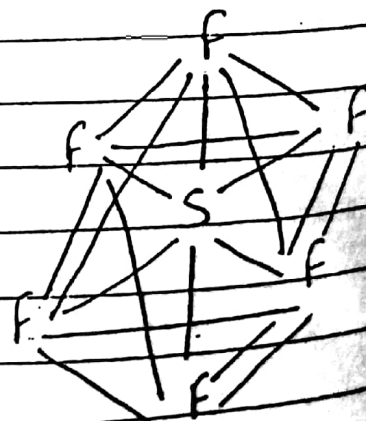
4.) ~~3.)~~ SF_6



In presence of F, e get excite to 3d.



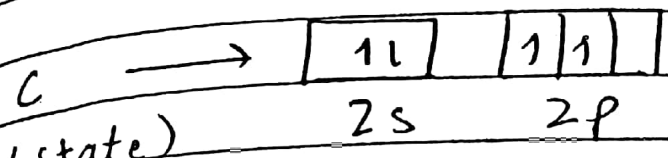
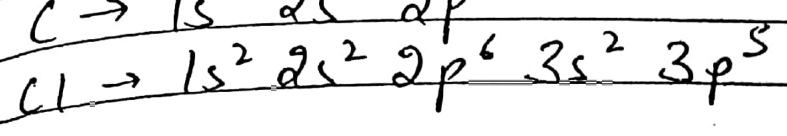
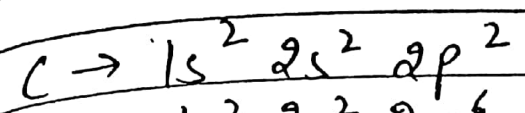
or



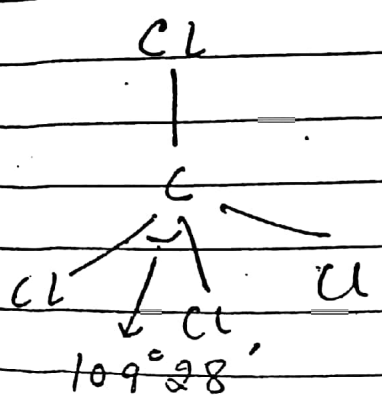
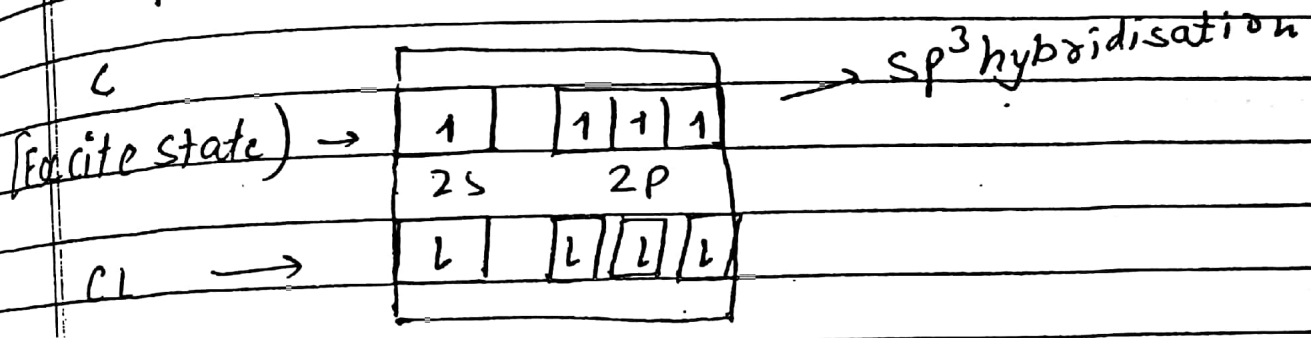
Shape and geometry $\rightarrow$ Octahedral.

The shape and geometry is same due to absence of lone pair.

5.) CCl_4

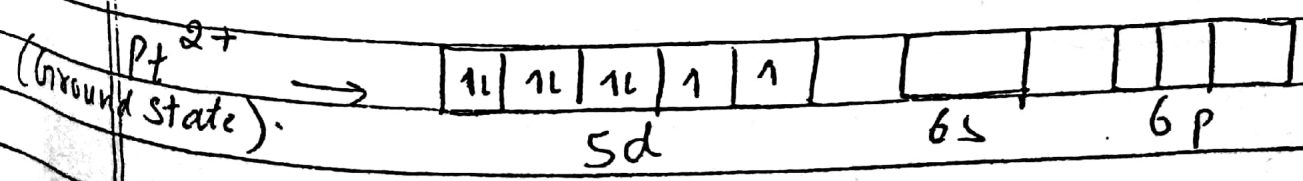
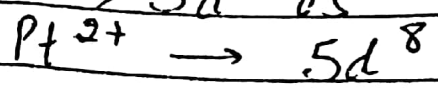
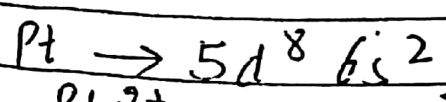


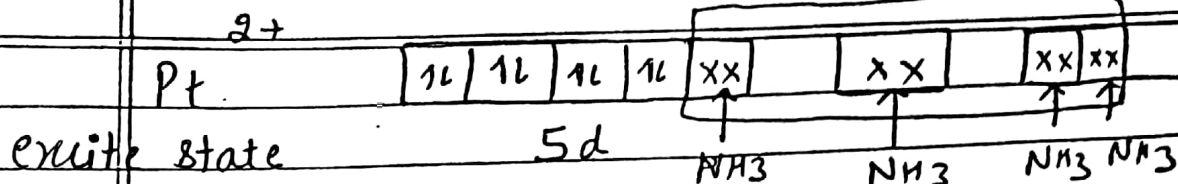
(Ground state) In presence of Cl, e^- get excited to 2p from 2s-



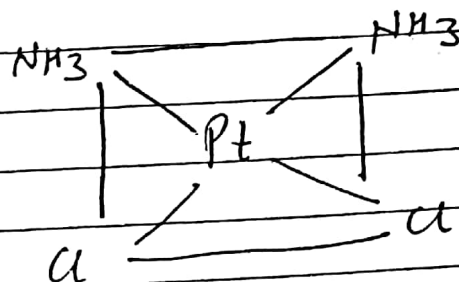
Geometry $\rightarrow$ Tetrahedral
Shape $\rightarrow$ Tetrahedral

6.) $[Pt(NH_3)_2 Cl_2]$





dsp^2 hybridisation
 Shape $\rightarrow$ square planar
 Geometry $\rightarrow$ square planar



CONTINUE

* FACTOR ELECTRON AFFINITY $\rightarrow$

Exceptions $\rightarrow$

$\rightarrow$ As we move left to right, electron affinity inc.
 But Noble gas has zero electron affinity but because due to inert or stable electronic configuration

$\rightarrow Be, Mg, N, P \rightarrow$ stable electronic configuration $\rightarrow$ zero or very low E.A.

$\rightarrow$ Halogens ($ns^2 np^5$) $\rightarrow$ Highest E.A.

$\rightarrow$ As we move Top to bottom in halogens, E.A. dec. due to which E.A. of F is more than Cl. But it can't be happen. Actually, E.A. of F is less than E.A. of Cl due to that reason $\rightarrow$

(i) F has small size due to which electron-electron repulsion will take place. If e^- is added in it. whereas Cl is large in size and has large

3p subshell due to which it can gain e^- easily.

ELECTRONEGATIVITY $\rightarrow$

It is the tendency of an atom to attract the shared pair of an e^- in a molecule.

For e.g $\rightarrow H : Cl$

In $H : Cl$, the whole pair of e^- is attracted by Cl , so, Cl is more electronegative.

$Cl : F$

In $Cl : F$, the whole pair of e^- is attracted by F , so, F is more electronegative.

★ FACTORS AFFECTING ELECTRONEGATIVITY $\rightarrow$

1) Size of an atom $\rightarrow$

If $r \downarrow \rightarrow F.A. \uparrow \rightarrow E.N. \uparrow$

2) Oxidation state $\rightarrow$

If +ve O.S. $\uparrow \rightarrow r \downarrow \rightarrow F.A. \uparrow \rightarrow E.N. \uparrow$

If -ve O.S. $\uparrow \rightarrow r \uparrow \rightarrow F.A. \downarrow \rightarrow E.N. \downarrow$

3) Type of hybridisation $\rightarrow$

s character

CH_4 sp^3 25%

$CH_2=CH_2$ sp^2 33%

$CH \equiv CH$ sp 50%

If s character $\uparrow \rightarrow E.N. \uparrow$

4) Bond order $\rightarrow$

If Bond order $\uparrow \rightarrow E.N. \uparrow$

5) Nature of Substituents $\rightarrow$

$E.N.$ of Cl in $H-Cl > E.N.$ of Cl in $Cl-F$

$\delta^+ \text{H} : \delta^- \text{Cl} \rightarrow \text{polar covalent bond}$

DATE: / / 20
PAGE No

★ VARIATIONS IN PERIODS $\rightarrow$

From left to right $\rightarrow$

$\chi \downarrow \rightarrow \text{F.A.} \uparrow \rightarrow \text{F.N.} \uparrow$

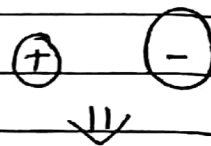
★ VARIATIONS IN GROUPS $\rightarrow$

From Top to bottom $\rightarrow$

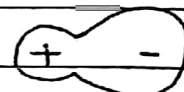
If $\chi \uparrow \rightarrow \text{F.A.} \downarrow \rightarrow \text{F.N.} \downarrow$

POLARISING POWER $\rightarrow$

It is the tendency to attract the electron cloud towards itself.



When a +vely charge ion come close together, then e^- cloud of cation is attracted towards anion. Similarly, electron cloud of anion is attracted towards cation. But cation being small in size has greater tendency to attract the e^- cloud of anion towards itself.



Power of cation to polarize the electron cloud of anion towards itself is called Polarizing power.

Tendency of electron cloud of anion to get polarized towards cation is called Polarizability.

★ FACTORS AFFECTING POLARIZING POWER $\rightarrow$ (Fajan Rule)

- (I) If size of cation $\downarrow \rightarrow$ Polarizing power $\uparrow$
- (II) If size of anion $\uparrow \rightarrow$ Polarizability $\uparrow$
- (III) If charge on cation $\uparrow \rightarrow$ size $\downarrow \rightarrow$ Polarizing power $\uparrow$

(iv) If charge on anion $\uparrow \rightarrow$ Size $\uparrow \rightarrow$ Polarizability $\uparrow$

OXIDATION STATE $\rightarrow$

It is the no. of electron gain or loss, is called oxidation state.

For s and p group element $\rightarrow$

ns^1	ns^2	ns^2, np^1	ns^2, np^2	ns^2, np^3	ns^2, np^4	ns^2, np^5
Grp. 1	Grp. 2	Grp. 13	Grp. 14	Grp. 15	Grp. 16	Grp. 17

Common O.S.	+1	+2	+3	+4	-3	-2	-1
			+1 (In, Th) (inert pair effect) (because ns^2 take part in bond formation but ns^2 does not take)	+2 (Pb)	+5 (P, Os)		

For d-group element $\rightarrow$

$\rightarrow$ 1st transition series $\rightarrow$

→ 1st transition series →									
		$3d^3 4s^2$		$3d^5 4s^2$		$3d^7 4s^2$		$3d^{10} 4s^1$	
$3d^1 4s^2$	$3d^2 4s^2$	↑	$3d^4 4s^1$	↑	$3d^6 4s^2$	↑	$3d^8 4s^2$	↑	$3d^{11} 4s^2$
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
+2	+2	+2	+1	+2	+2	+2	+2	+1	+2
+3	+3	+3	+2	+3	+3	+3	+3	+2	
	+4	+4	+3	+4		+4			
		+5	+4	+5					
			+5	+6					
			+6	+7					

CO-ORDINATION NUMBER $\rightarrow$

It is the no. of ligands surrounding the central metal ion, is called co-ordination number.

$\rightarrow$ Co-ordination No. vary from 2 to 9.

acid $\rightarrow$ e^- acceptor

Base $\rightarrow$ e^- donor

	C.N.	Shape
for e.g. $\rightarrow$ (i) $[Ag(NH_3)_2]^+$	2	linear
(ii) $[Ni(CN)_4]^{2-}$	4	square planar
(iii) $[Cr(H_2O)_6]^{2+}$	6	octahedral

* HARD AND SOFT ACID AND BASES $\rightarrow$ (Pearson concept)

It can be explained by Pearson concept.
In metal complex:

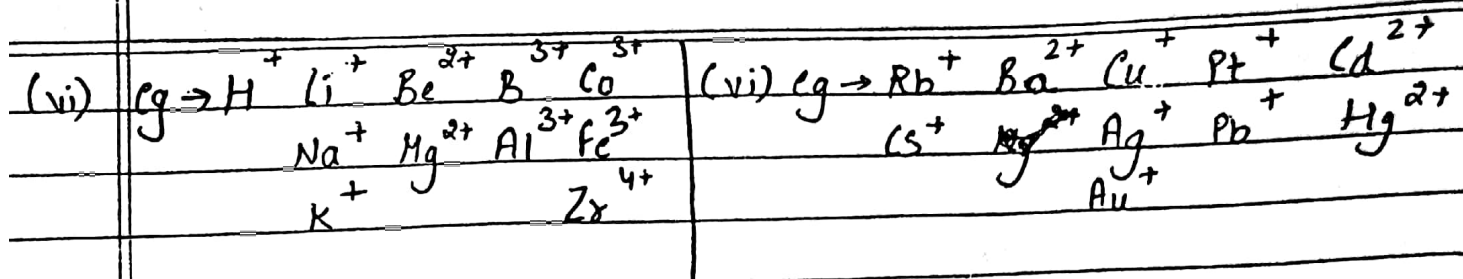
Central metal ion (Lewis acid) $\begin{cases} \text{Hard acid} \\ \text{soft acid} \end{cases}$

Central metal ions known as Lewis acid bcz central metal ion carry +ve charge and it accept e^- pair from ligand.

Ligand (Lewis base) $\begin{cases} \text{Hard base} \\ \text{soft base} \end{cases}$
(-vely charged or neutral)

Ligand is known as Lewis base bcz ligands donate e^- pair.

Hard acid	Soft acid
(i) Small size metal ion	(i) larger size metal ion
(ii) Electronegativity larger	(ii) Electronegativity smaller
(iii) Electron cloud is not easily polarized.	(iii) Electron cloud can easily be polarized.
(iv) lighter metal ion	(iv) Heavier metal ion
(v) It has generally higher O.S.	(v) It has zero or low oxidation state.



Hard base	Soft base
(i) Small Size	(i) larger size
(ii) E.N. ↑	(ii) E.N. ↓
(iii) Electron cloud is not easily polarized	(iii) Electron cloud can easily be polarized.
eg → N, O, F, Cl NH_3 O^{2-} F^- Cl^- ROH CO_3^{2-}	eg → P, S, I PH_3 S^{2-} I^- PO_4^{3-} R_2S H_2S

HARD SOFT ACID BASE (HSAB) PRINCIPLE →

Status Hard acid prefer to combine with hard base and Soft acid prefer to combine with soft base.

for e.g → $NaCl$ exist, $CaCO_3$ exist, NaI does not exist bcz Na^+ is Hard acid but I^- is Soft base.

APPLICATIONS →

1. To predict stability of complex.
for e.g → AgI_2^- and AgF_2^-
 SA SB SA HB

As, AgI_2^- is more stable than AgF_2^- bcz soft acid combine with soft base.

donor side.
e.g. $\rightarrow \text{SCN}^-$, NCS , CN^- , NC^-

(ii) $[\text{CoF}_6]^{3-}$ is more stable than $[\text{CoI}_6]^{3-}$
HA HB HA SB

2. To predict the formation of complex with ambident ligand.

For e.g. $\rightarrow [\text{Co}(\text{NCS})_4]^- \Rightarrow \text{Co}$ is linked with N atom
HA HB

$[\text{Pd}(\text{SCN})_4]^- \Rightarrow \text{Pd}$ is linked with S atom
SA SB

Here, Pd linked with S but not with N bcz
Pd is Soft acid and S is soft base.

3. To predict the feasibility of rxn,

for e.g. $\text{LiI} + \text{CsF} \rightarrow \text{LiF} + \text{CsI} \rightarrow \text{Feasible}$
HA SB SA HB

As combination of LiI and CsF is not stable bcz in which if one is Hard acid then other is soft base i.e. not perfect combination. So, this rxn is feasible

$\text{BeI}_2 + \text{HgF}_2 \rightarrow \text{BeF}_2 + \text{HgI}_2 \rightarrow \text{Feasible}$
HA SB SA HB

4. To predict the occurrence of Oxe

for e.g. $\rightarrow \text{NaCl}$, CaCO_3 , MgCO_3 , Al_2O_3 , HgS , Ag_2S
(HA HB) (HA HB) (HA HB) (HA HB) (SA SB) (SA SB)

5. HA-HB combine to form ionic bond.

for e.g. $\rightarrow \text{NaCl}$, LiF

SA-SB combine to form covalent bond

for e.g. $\rightarrow \text{HgI}_2$